

Exact Product-Kernel Gelation and the Smoluchowski–Flory Postgel Boundary

HCS Research Program

29 August 2026 (revision 0)

Abstract

We give a convention-complete solution atlas for monodisperse discrete coagulation with multiplicative kernel. Cayley tree coefficients close every cluster concentration before gelation, while rooted and unrooted tree functions give the mass and number generating functions. The second moment diverges at $t = 1$, where the concentrations have the critical $k^{-5/2}$ tail. We then resolve a postgel ambiguity explicitly. With the Smoluchowski finite-sol loss, one Stockmayer continuation freezes the critical shape and has sol mass $1/t$. With gel-reactive Flory loss, the pregel coefficient formula continues and has sol mass $q = -W_0(-te^{-t})/t$. The two families agree at $t = 1$ but direct substitution shows that they solve different equations afterward. No universal weak-solution uniqueness is claimed. Exact independent reconstruction, symbolic identities, canonical replay, and hostile mutations audit the theorem.

1 Frozen kinetic convention

For $k \geq 1$, set

$$\dot{c}_k = \frac{1}{2} \sum_{i+j=k} ij c_i c_j - k c_k M_1(t), \quad M_q(t) = \sum_{k \geq 1} k^q c_k(t), \quad (1)$$

with $c_1(0) = 1$ and $c_k(0) = 0$ for $k > 1$. Thus $K(i, j) = ij$, the unordered gain has factor $1/2$, and (1) uses only finite-cluster mass in its loss. A different gel-reactive loss will be introduced only after its convention is named.

Define the rooted and unrooted labelled-tree series

$$T(u) = \sum_{k \geq 1} \frac{k^{k-1}}{k!} u^k, \quad U(u) = \sum_{k \geq 1} \frac{k^{k-2}}{k!} u^k. \quad (2)$$

Cayley's formula and Lagrange inversion give $T = ue^T$, $U = T - T^2/2$, and $uU'(u) = T(u)$.

Theorem 1 (Exact pregel solution and gel point). *For $0 \leq t \leq 1$,*

$$c_k(t) = a_k t^{k-1} e^{-kt}, \quad a_k = \frac{k^{k-2}}{k!}. \quad (3)$$

If $u = tze^{-t}$, the mass and number generating functions are

$$G(z, t) = \sum_{k \geq 1} k c_k z^k = \frac{T(u)}{t}, \quad C(z, t) = \sum_{k \geq 1} c_k z^k = \frac{U(u)}{t}, \quad (4)$$

as formal power series in z , and analytically on the principal convergence disk (in particular $0 \leq z \leq 1$), with continuous $t = 0$ values. For $t < 1$,

$$M_0 = 1 - \frac{t}{2}, \quad M_1 = 1, \quad M_2 = \frac{1}{1-t}, \quad M_3 = \frac{1}{(1-t)^3}. \quad (5)$$

At $t_g = 1$,

$$c_k(1) \sim \frac{1}{\sqrt{2\pi}} k^{-5/2}, \quad (6)$$

so M_2 diverges. For fixed $0 < t < 1$,

$$c_k(t) \sim \frac{k^{-5/2}}{\sqrt{2\pi} t} [te^{1-t}]^k. \quad (7)$$

Proof. Coefficient extraction from $T = ue^T$, or the standard labelled-tree edge decomposition, gives

$$(k-1)a_k = \frac{1}{2} \sum_{i+j=k} ija_i a_j. \quad (8)$$

At $t = 0$, equation (3) means its continuous limit: $c_1 = 1$ and $c_k = 0$ for $k > 1$, so no 0^0 convention is left implicit. Differentiating (3), using (8), and imposing $M_1 = 1$ proves (1) coefficient by coefficient. Substitution in (2) gives (4). On the branch reached from zero, $T(te^{-t}) = t$ for $t \leq 1$, so $U = t - t^2/2$. Applying $z\partial_z$ to G and using $uT'(u) = T/(1-T)$ gives (5); equivalently $\dot{M}_2 = M_2^2$ and $\dot{M}_3 = 3M_2M_3$. Stirling's formula applied to (3) gives (6) and (7), and (6) makes $\sum k^2 c_k$ divergent. $\square$

References

- [1] J. B. McLeod, *On an Infinite Set of Non-Linear Differential Equations*, Quarterly Journal of Mathematics 13 (1962), 119–128. DOI: 10.1093/qmath/13.1.119.
- [2] J. B. McLeod, *On an Infinite Set of Non-Linear Differential Equations, II*, Quarterly Journal of Mathematics 13 (1962), 193–205. DOI: 10.1093/qmath/13.1.193.
- [3] R. M. Ziff and G. Stell, *Kinetics of polymer gelation*, Journal of Chemical Physics 73(7) (1980), 3492–3499. DOI: 10.1063/1.440502.
- [4] C. Normand and L. Zambotti, *Uniqueness of post-gelation solutions of a class of coagulation equations*, Annales de l'Institut Henri Poincaré C 28(2) (2011), 189–215. DOI: 10.1016/j.anihpc.2010.10.005.
- [5] J. R. Norris, *Smoluchowski's coagulation equation: uniqueness, nonuniqueness and a hydrodynamic limit for the stochastic coalescent*, Annals of Applied Probability 9(1) (1999), 78–109. DOI: 10.1214/aoap/1029962598.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero, universal postgel-uniqueness, or Hilbert–Pólya claim. **Data and code.** The exact coefficient ledger and all audits accompany HCS-C228. **AI-use disclosure.** Generative tools assisted drafting and code generation; the released internal chain checks the displayed formulas and metadata. This is not external peer review.